

L3: Lesson Summary



The main objective of this lesson was to explore the notion of “**degree distribution**” for a given network. The degree distribution is probably the first thing you will want to see for any network you encounter from now on. It gives you a quantitative and concise description of the network’s connectivity in terms of average node degree, degree variability, common degree modes, presence of nodes with very high degrees, etc.

In this context, we also examined a number of related topics. First, the friendship paradox is an interesting example to illustrate the importance of degree variability. We also saw how the friendship paradox is applied in practice in vaccination strategies.

We also introduced $G(n,p)$, which is a fundamental model of random graphs – and something that we will use extensively as a baseline network from now on. We explained why the degree distribution of $G(n,p)$ networks can be approximated with the Poisson distribution, and analyzed mathematically the size of the largest connected component in such networks.

Obviously, the degree distribution does not tell the whole story about a network. For instance, we talked about networks with degree correlations. This is an important property that we cannot infer just by looking at the degree distribution. Instead, it requires us to think about the probability that two nodes are connected as a function of their degrees.

We will return to all of these concepts and refine them later in the course.